Class 2

- $N = \{1, 2, 3\}$
- Each player chooses a number between $[0, 100]$
- $2/3$ of avg is calculated
- The player who chooses closest to $2/3$ of the avg is the winner.

Outcomes are

- Player 1 is the sole winner
- " 2 " " " "
- " 3 " " " "
- Player 1 and 2 are joint winners
- Player 1 and 3 are joint winners
- Player 2 and 3 are joint winners
- All are winners

} 0

Player 1's preference over the outcomes

Player 1 is the sole winner (1)

$\succ_1$

Player 1 and 2 are joint winners $\sim_1$ Players 1 and 3 are joint winners $\frac{1}{2}$

$\succ_1$

All are winners $1/3$

$\succ_1$

all the other outcomes 0

$\succeq_1 \subseteq O \times O$

(outcome x, outcome y) $\in O \times O$

Player 1 weakly prefers outcome x over outcome y.

A utility function Suppose O is the set of outcomes. Then a utility function u is a mapping from O to $\mathbb{R}$.

A representation of a preference over O

A utility function u represents a preference $\succsim$ over O if $\forall x, y \in O$

$$x \succsim y \iff u(x) \geq u(y)$$

Theorem Suppose O is finite. Then a preference can be represented by a utility function iff the preference is complete and transitive.

If O is infinite, a similar result holds with some more assumptions of continuity.

Normal form / Strategic form games

A normal form game is a 3-tuple

$$\langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle \text{ where}$$

- N is a finite set of players
- S_i: set of strategies for player i
- $u_i : \times S_i \to \mathbb{R}$

$i \in N$

For the beauty contest games

- $N = \{1, 2, 3\}$
- $S_1 = [0, 100]$ $S_2 = [0, 100]$ $S_3 = [0, 100]$
- $$u_i(x_1, x_2, x_3) = \begin{cases} \frac{1}{|W(x_1, x_2, x_3)|} & \text{if } i \in W(x_1, x_2, x_3) \\ 0 & \text{o.t.} \end{cases}$$

$$W(x_1, x_2, x_3) = \arg\min_{i \in N} \left| x_i - \frac{2}{3}\bar{x} \right|$$

Prisoner's Dilemma

- $N = \{P_1, P_2\}$
- $S_1 = S_2 = \{C, D\}$

P_1 \ P_2	C	D
C	(-1, -1)	(-5, 0)
D	(0, -5)	(-3, -3)

In this game players will play (D, D) as for both of them D dominates C.

Battle of sexes

- $N = \{$man, woman$\}$
- $S_M = S_W = \{F, D\}$

M \ W	F	D
F	(2, 1)	(0, 0)

D	(0,0)	(1,2)